

An investigation into existing and potential economic impacts of internet restrictions

Abstract

This paper investigates the extent to which internet restrictions are deployed within India, Sudan, Bangladesh, and China or are strongly suspected to be planned in Russia. For each country, the severity, nature, and duration of the internet restrictions, as well as the reason for their existence, is evaluated. Using this information, a holistic interpretation of the economic impact is distilled. The impacted areas include not only the national level but also down to a personal level. Russia seeks to create an intranet whereby its citizens will only be able to access Russian websites. This can have both positive and negative impacts on its economy, which is nuanced due to the war in Ukraine. Using the Sudan case study, an evaluation using the COST tool can expose both the tool's strengths and limitations in aiding this investigation. In Bangladesh, the unlikely culprit that has devastated its economy in the face of an internet blackout is the garment industry, as well the ramifications on its telecommunications sector. In China, these restrictions can differ from region to region and have varying effects on the economy as they can act both as a protectionist policy and as an inhibitor of growth.

Introduction

Internet restrictions fall under two categories. Government interventions can restrict the access to the internet that their citizens benefit from, and foreign intervention including sabotage or cyberwarfare. This essay explores the potential economic impact of internet restrictions with a focus on the former category.

Today, the internet is under threat. The internet serves as a lifeline for the freedom of expression, which threatens some countries with authoritarian regimes putting it into the crosshairs of those regimes, leading to extreme measures being prepared by Russia¹ for future stifling of unrest, or the Indian suppression of unrest in the Kashmir region². Intentional outages are increasing in length of time and frequency. For example, in 2024, there was an alarming increase of 12% in the length of time³ of deliberate, government-sanctioned internet restrictions.

¹ Porter, Tom. "Russia Tested Cutting off Internet Access: Report." Business Insider, December 12, 2024. <https://www.businessinsider.com/russia-tested-cutting-off-internet-access-report-kremlin-runet-2024-12>.

² Mulani, Nikhil. "The Politics of Internet Shutdowns: Governance Lessons from Kashmir's Internet Shutdowns." *Verfassungsblog*, December 20, 2022. <https://doi.org/10.17176/20221221-001724-0>.

³ Woodhams, Samuel, and Simon Migliano. "The Global Cost of Internet Shutdowns." *www.top10vpn.com*, May 8, 2024. <https://www.top10vpn.com/research/cost-of-internet-shutdowns/>.

What is also perhaps less evident but just as important, is the economic aspect of this issue as the victim of these measures is the lifeline of many lives. As the potential uses of the internet continue to expand and more and more businesses pop up online, the impact of the internet grows. The advent of exclusively online businesses that have become household names, such as Amazon, which brings in an annual revenue of \$574,780,000,000⁴, have only reinforced the need to protect people's access to the internet and enshrine it as a fundamental right, whilst protecting it from saboteurs seeking to destabilize countries.

To make the best use of available information, case studies are examined systematically by employing three research questions. Firstly, the extent to which the internet is restricted - which can range from how many people are affected to the amount of industries impacted. Then, the nature of these restrictions must be analysed as the cause for their enactment is crucial to understanding future impact as a result of recurrence. Lastly, the severity of the restrictions - whether they represent full blackouts or only some services are restricted or perhaps throttling is used. The extent to which the global economy is affected by each of these outages must also be examined. Unfortunately there are far too many potential case studies to examine them all, but using a selection from several continents spanning different economic contexts, this paper should demonstrate the real and quantifiable impact. This can be accomplished by using pre-established methods which already exist as a result of the efforts by dedicated NGOs - such as the Advocacy assembly⁵ or the #KeepitOn coalition, which is made up of many other organizations all working together specifically to fight against intentional outages. These methods will be used to analyse impacts, which in turn will be applied to those case studies and presented in a digestible manner. The countries where case studies for present and potential restrictions to be evaluated include Russia, China, India, Sudan, and Bangladesh.

Russia

Russia is well-known for its authoritarian government under President Vladimir Putin who has displayed aggressive behaviour towards the west, such as the invasions of Ukraine in 2014 and the ongoing conflict since 2022. The war therefore has reinforced the motivation Russia has to spread misinformation and reduce the access to foreign media outlets, which may provide information regarding the war sanctioned by the government which may counter their narrative.

⁴ Slingerland, Cody. "The Cost of Shutting down the Internet." CloudZero, July 15, 2024.

<https://www.cloudzero.com/blog/cost-of-shutting-down-the-internet/>.

⁵ Assembly, Advocacy. "Measuring the Economic Costs of Internet Shutdowns." Accessed November 4, 2024.

https://advocacyassembly.org/documents/32/EN_Designed_Toolkit_Unveiling_the_Economic_Costs_of_Internet_Shutdowns.pdf.

In order to suppress their internal unrest, they have tested a way by which they can continue having a network and yet isolate themselves from the internet. In short, they have successfully physically separated entire regions from the global internet.⁶ The nature of those regions - unstable - reveal the actual intention. Regardless, this is a major difference between the restrictions in China and this new system. Whereas the Chinese system can be bypassed by people in most regions using a VPN, the Russian implementation would prove to be definitive in its capacity to separate RuNet from the world; it involves a physical separation from the world internet, preventing citizens from accessing foreign companies' online services.

Right now, the Russian government denies any such plans. But if history is anything to go by, that is not to be trusted on this topic, especially as it has recently tested capabilities which accomplish those very goals, even if they had denied working on this as recently as 2022.⁷ Therefore, without any evidence to support the contrary, the trend is that they will continue developing this technology and working to implement it despite any future claims they may make. The next logical question is, what kind of economical impact will follow for Russia and its citizens if it were to introduce this nation-wide. Russia has always been more isolated from its neighbours as opposed to the other countries mentioned in this essay. One such reason has been Russophilia which involves admiration and fondness of Russia. Many services where western countries would depend on each other such as Facebook, Google etc have equivalent national services such as Vk, which would still be operational. Russia is less dependent on its own neighbours especially having suffered sanctions following its Ukrainian conflict which have already dealt massive blows to any incentives for international trade.

Therefore, the question for Russia of to what extent its economy would be impacted should rather be to what extent would what's left of its economy be impacted. The answer to this question must be answered by referring to the other case studies. Key sectors such as international tourism have much to suffer. They would no longer be able to post ads for their respective destinations and communication with foreign customers would be eliminated.⁸ In this case, slammed with so many sanctions, and with virtually most western governments strongly warning their citizens against travelling to Russia, this factor is not likely to make the same dent as it would have prior to these tensions.

On the other hand, it may not be so simple as Russia has a growing domestic and international tourism market. For example, Russian hotel bookings grew by 16.1% in 2024 compared to 2023. Similarly, domestic air traffic also rose 5% in terms of sales. Foreign visitors - especially the

⁶ Porter, Tom. "Russia Tested Cutting off Internet Access: Report." Business Insider, December 12, 2024. <https://www.businessinsider.com/russia-tested-cutting-off-internet-access-report-kremlin-runet-2024-12>.

⁷ Meyer, David. "For a Country Denying It Will Quit the Internet, Russia's Acting a Lot like It." Fortune, March 7, 2022. <https://fortune.com/2022/03/07/russia-runet-disconnect-ukraine-dns-chernenko-letter/>.

⁸ Mitchell, Robert. "The Real Impact of Internet Shutdowns." Internet Society, June 28, 2023. <https://www.internetsociety.org/blog/2023/06/the-real-impact-of-internet-shutdowns/>.

Chinese people also make up a significant portion of the tourism sector. This means that the Russian tourism sector continues to be very important and has a lot of potential that currently goes unexploited. Even as domestic tourism grows, the increase in outbound tourism has reached 20% - encompassing about 29-30 million people.⁹ If Russia were to isolate its internet, it is possible that this would annihilate foreign offerings and drive up demand for their local alternatives. At the same time however, there is still a negative pressure on the domestic economy as it would prevent Russian businesses from advertising to foreigners.

If Russia would face full internet blackout, however, according to the COST tool, \$12,140,458,146 would be the total damages to the economy given it would be a blackout for 30 days. Despite these figures, as it stands now, the direct economic costs of isolating the Russian economy don't seem to be so dire. Regardless, in the long run, this does have the risk to incentivise the inefficiency of Russian companies due to the much weaker competition they'll face. This may determine a serious lack of innovation as they'll be shielded from competitors - in line with other protectionist schemes which have already been proven to yield stagnation. On the other hand, as a protectionist policy, it could foster a safe haven for Russian start-ups and encourage users to migrate to national services. This may help to reduce the capital repatriated abroad without impacting users, leading to an increase in the Russian economy. China is in a similar position here, whose great firewall appears to have had a positive impact on its domestic industries with companies such as Alibaba or Tencent being listed as some of the richest companies in the world. However, those countries do interact significantly with foreign markets, something which would be decimated under the Russian solution, so it is possible that such growths would not be observed.

Sudan

Unlike Russia which has a strong government in power, Sudan is currently experiencing a brutal civil war. One of the casualties of this conflict is the internet. Both the Rapid Support Forces (RSF) and the Sudanese Armed Forces (SAF) have weaponized internet shutdowns in order to gain the upper edge. The shutdowns began on the 2nd of February 2024, and lasted a month, as a result of the seizure of the internet service providers (ISPs), who are responsible for the distribution of the internet throughout Sudan. Unfortunately this has had dire consequences for the already suffering Sudanese economy. Wiring money or using e-wallets, something that is widely used in Sudan which is important for the conduct of businesses as without them

⁹ Galia Garifullina. "Russia's Domestic Tourism Grows as Hotel Prices Rise by 16-21%." Реальное время. PLC "Realnoe Vremya," December 20, 2024.
<https://realnoevremya.com/articles/8438-russias-domestic-tourism-grows-as-hotel-prices-rise-by-16-21>.

transactions can either be completely impossible or far less convenient (which also leads to lower demand of services by potential customers).¹⁰

The COST tool, which uses economic indicators from 2020¹¹ is based on the Brookings Institution method. It works by taking the National GDP, multiplying it by the percentage of the year for which the internet has been shut down, multiplied by the portion that the economy is actually digital, and adds the multiplier effect of the disrupted digital economy.¹² At the time of writing, for a duration of 30 days, using a digital economy extent of 0.03, affecting all internet services including landline and mobile services, the estimated national economic impact is SDG 103,849,854,156, or USD 172,795,098. As it is only a partial blackout however, and isn't necessarily continuous, the actual damage is only a portion of the aforementioned value which would be true for a full blackout. When taking into account the nature, duration and severity, the value yields only a loss of 12.4 million US dollars.¹³ Regardless, this is still by no means a small amount of money.

As Sudan suffers from harsh poverty, with a population of 48 million and over nine million internally displaced¹⁴ all resources must be directed towards easing that burden. 4.8 million people are directly affected by these outages - in total lasting for 212 hours. Instead, by squandering this money, businesses dry up in terms of revenue and scale down their operations in the best case scenario. In the worst case scenario, businesses lose their entire model of business. In both cases, irreversible economic damage is done, with structural unemployment rising significantly, all with an unstable government unable to provide economic relief and welfare to victims. Sudan's digital sector is a growing, young sector which needs to be nurtured especially as it represents the future. It is part of the tertiary sector - the services sector, which is the pinnacle of economic development and is where Sudan should invest in the most. By forsaking the sector, Sudan faces a massive economic setback which is the last thing it needs right now.

Iran

In contrast to Sudan which has an economy of only 109.3B, Iran's GDP stands at 404.6B, thereby meaning there is a larger amount at stake.¹⁵ Iran experienced a 17 month long social

¹⁰ KeepItOn. "#KeepItOn in Times of War: Sudan's Communications Shutdown Must Be Reversed Urgently - Access Now." Access Now, February 13, 2024. <https://www.accessnow.org/press-release/keepiton-sudan-shutdown/>.

¹¹ Cost of Shutdown Tool," netblocks.org, n.d., <https://netblocks.org/cost/>

¹² M. West, Darrell. "Internet Shutdowns Cost Countries \$2.4 Billion Last Year." Brookings, October, 2016. <https://www.brookings.edu/wp-content/uploads/2016/10/internet-shutdowns-v-3.pdf>

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¹⁴ KeepItOn. "#KeepItOn in Times of War: Sudan's Communications Shutdown Must Be Reversed Urgently - Access Now." Access Now, February 13, 2024. <https://www.accessnow.org/press-release/keepiton-sudan-shutdown/>.

¹⁵ World Bank. "World Bank Group - International Development, Poverty and Sustainability." Worldbank.org, 2023. <https://www.worldbank.org/ext/en/home>.

media ban with occasional full internet shutdowns which caused 1.6 billion USD in damages to the Iranian economy, which started in September of 2022. The internet was down for a total of 7171 hours in 2022 costing the country US\$773 Million. The ISNA news agency reported that every hour without the internet cost Iran US\$1.5 million. At the end of the 17 month shutdown, a total of 1.6 billion USD was incurred as damage to the Iranian economy as a direct consequence of the restrictions.¹⁶ In 2023, Iran's internet was down for 13,140 hours. This figure is split between a social media shutdown amounting to 12624 hours and a full internet blackout which lasted 516 hours total. This cost the country 920.3 million dollars and directly affected 33.7 million people.¹⁷

A very affected segment was that of online shopping and electronic payment processing. 90 percent of transactions occurred online prior to the protests which spurred the internet shutdowns. For example, two Iranian companies which provided electronic payment services for businesses, Zibal and Zarin Pal, saw a respective 55 and 60 percent drop in the amount of transfers they processed. The Instagram ban alone led to the shutting down of more than 500 thousand Iranian small businesses, having a direct impact on one million people with indirect impacts on 8 million people.¹⁸ This is bound to have had a detrimental impact particularly on the low-income part of the population as it would have pushed more people into that bracket. Moreover, with the sudden structural unemployment as the result of a huge amount of businesses, the sudden influx of available workers would have made jobs very scarce as each would have much more competition, threatening those in different sectors. This impact is also not to be underestimated, as that 1 million people who suddenly no longer had an income found themselves unable to purchase any sort of products, leading to lost revenue not only for the government which would then also need to fund the welfare of the suddenly unemployed population, but also for the other businesses which provided services and products to them. This causes a domino effect which pushes down on every sector of the economy and every citizen. This is felt economically by the whole country, not just a subset of it as the whole economy is interconnected.

Bangladesh

Bangladesh has a slightly larger economy than Iran at 437.4B. It lost 1.2 billion US dollars in revenue over the span of five days in 2024 as a result of its garment export sector being heavily

¹⁶ Iran, A. correspondent in. "17 Months of Internet Shutdown Costs Iran Billions." <https://www.newarab.com/>, February 14, 2024.

<https://www.newarab.com/news/17-months-internet-shutdown-costs-iran-billions>

¹⁷ Woodhams, Samuel, and Simon Migliano. "The Global Cost of Internet Shutdowns." [www.top10vpn.com](https://www.top10vpn.com/research/cost-of-internet-shutdowns/), May 8, 2024. <https://www.top10vpn.com/research/cost-of-internet-shutdowns/>.

¹⁸ Iran, A. correspondent in. "17 Months of Internet Shutdown Costs Iran Billions." <https://www.newarab.com/>, February 14, 2024. <https://www.newarab.com/news/17-months-internet-shutdown-costs-iran-billions>

affected by the internet blackout which has battered its economy. It has completely shut down e-commerce, and f-commerce - facebook - based markets. These sectors are accumulating 5 million US dollars every day in losses of revenue.¹⁹ According to the COST tool²⁰, \$393,307,644 is the sum of money Bangladesh would have lost. The variables automatically selected by the tool are “Using approximated digital economy extent of 0.03” and the economic indicators of 2020. There is a massive discrepancy between the 1.2 billion dollars which come from just one sector and the tool which calculates the only the digital economy impact, however this stems from the fact that the tool takes into account just the digital sector whereas here the garment sector is the biggest victim despite not being a traditionally digital sector. This just goes to prove that even whilst this sector is a huge victim, it is not often thought of in this context and yet has an enormous economic impact. Therefore in this circumstance the increasing reliance of the garment sector of the economy on the internet is now evident. In other cases yet to happen, the risks are also there, but may be just as well hidden throughout other industries. Even though it may not yet seem obvious, the burden will differ from country to country but will be expensive.

This ban has also affected people at a much lower level; people had grown accustomed to using services which offered them the convenience of being able to purchase most items including groceries online. This has become a chore now to a huge swathe of the population, leading to reduced demand as a result of the increased burden on their part, and the crippling of the services that provided them with this ability. If the internet is not restored, not only is the loss in revenue catastrophic, structural unemployment will occur once those firms become insolvent, as with the Iranian case, and with the earlier hidden vulnerability which was the garment sector, a domino effect will ravage the economy further bringing about a harsh economic recession.

Likewise, call centers also pose the same risks, with them having lost 3 million dollars a day during the shutdown. Two-thirds of the operations of call-centers are with global companies. Not only does this mean that the internet shutdown represents a complete cease of operations due to the way the companies operate, as coordinating what is mostly customer support without the internet is impossible, but a massive loss of reputation. What makes Bangladesh very attractive to global companies seeking to outsource their labour to it is the low labour price. Unfortunately there are many other countries that share that advantage. This sudden drop in the confidence level that Bangladesh-based call centers are able to honour their contracts makes companies far more likely to consider switching providers in the long-run, making it hard to quantify the economic loss except in the short run, however it is just as significant if not more. There are also huge losses for commercial banks not only in terms of lost traffic but also the costs they incur.

¹⁹ Mahmud, Faisal. “Bangladesh Curfews, Internet Blackout Batter Economy amid Quota Protests.” Al Jazeera, July 23, 2024.

<https://www.alfcra.com/news/2024/7/23/bangladesh-curfews-internet-blackout-batter-economy-amid-quota-protests>.

²⁰ netblocks.org. “Cost of Shutdown Tool,” n.d. <https://netblocks.org/cost/>.

Every day they are unable to process scheduled transactions those transactions accrue interest which the bank then has to pay out of its own pocket, which amount to millions of dollars²¹

India

India is the second largest country in terms of the size of its economy considered in this essay, with a GDP of 3.568 trillion. In India, the Kashmir and Jammu regions have been heavily affected by internet blackouts.²² This is different to other case studies presented here since, as it is implemented regionally. This means that the COST tool cannot be used here as it doesn't currently support just regions for analysis. The mobile 4G internet was effectively down for 500 days. If it wasn't completely blacked out, it was throttled to 2G speeds, rendering it useless for almost anything except SMS messages. Whilst during those 500 days landlines were temporarily relaxed, this is effectively null since only about 3%²³ of the Indian population have access to landline internet. The blackouts have been (illegally²⁴) used as a political tool in order to stifle peaceful protests. These blackouts have far-reaching consequences.

In Kashmir, the local economy heavily relies on tourism. Unfortunately the tourism industry heavily depends on the internet to communicate with clients and advertise their destinations. The general secretary of the Travel Association of Kashmir, said life with slow internet "was worse than hell" and had led to severe losses in the business." Online businesses also fought to stay afloat, many of which didn't succeed in their struggles due to the shutdown in 2019 which caused thousands of people to experience structural unemployment. Some notable local companies that closed down include KartFood which used to be a food-delivery company and Kashmiri box which distributed Kashmiri goods, such as saffron, handicrafts or clothing, which only further hurt the Kashmiri' economy²⁵ as the producers of those goods now miss out on the

²¹ Mahmud, Faisal. "Bangladesh Curfews, Internet Blackout Batter Economy amid Quota Protests." Al Jazeera, July 23, 2024. <https://www.alfcra.com/news/2024/7/23/bangladesh-curfews-internet-blackout-batter-economy-amid-quota-protests>.

²² Bajoria, Jayshree. "'No Internet Means No Work, No Pay, No Food.'" *Human Rights Watch*, June 14, 2023. <https://www.hrw.org/report/2023/06/14/no-internet-means-no-work-no-pay-no-food/internet-shutdowns-denying-access-basic>.

²³ Mulani, Nikhil. "The Politics of Internet Shutdowns: Governance Lessons from Kashmir's Internet Shutdowns." *Verfassungsblog*, December 20, 2022. <https://doi.org/10.17176/20221221-001724-0>.

²⁴ Bajoria, Jayshree. "'No Internet Means No Work, No Pay, No Food.'" *Human Rights Watch*, June 14, 2023.

²⁵ Bajoria, Jayshree. "'No Internet Means No Work, No Pay, No Food.'" *Human Rights Watch*, June 14, 2023.

demand which used to come through those platforms, and are therefore missing out on large swathes of potential revenue.

On a local level, a few years prior to the lockdowns starting with 2019, the internet provided people with a unique opportunity to create their own businesses. Small handcrafted items such as crocheted products would be put up on Instagram by women, however this model was similarly annihilated by the lockdowns. Internet content creators are also severely affected by these internet shutdowns. Not only does their livelihood completely disappear when the internet is turned off, but even when it does turn back on, due to the algorithms with which content is pushed to the masses, if the upload schedule is inconsistent - as in the case where internet is periodically disrupted - the creator is punished due to the system not suggesting the content to many viewers. In order to put the sheer economic consequences into perspective for the average person, in November 2019, due to the fact that in Jammu landline internet service was restored but not in Kashmir, thousands of Kashmiris boarded a train to Banihal to access internet services. There, they submitted forms and applications, customers filing taxes, conducting banking transactions, and submitted official documents, from driver's licenses to tenders. They boarded this train for just five minutes of internet access - that's the extent to which people put themselves through huge inconveniences just to access the internet.

Another huge problem specific to India but which is sure to expand to other countries as digitalization becomes more common is with the NREGA. It is an act which is supposed to provide vital income security for over 100 million households in rural areas by guaranteeing them employment for 100 days. Due to the lack of employment possibilities in rural areas, this is a lifeline for those people, and even empowers women as 58% of participants were women from socially and economically marginalized households.²⁶ As a result of the government's digitalization push, however, attendance checks and wage payments are conducted exclusively online. As NREGA workers are required to be photographed and geotagged twice a day, when the internet is down, they are completely unable to work. The system is designed against a spotless internet, but due to the government it is anything but, and those who suffer are the workers themselves, unable to be paid and unable to work.

China

China holds second place after the US in terms of GDP, at around 17.79 trillion USD. This significantly contributes to why the Great Chinese firewall is perhaps the most known current

<https://www.hrw.org/report/2023/06/14/no-internet-means-no-work-no-pay-no-food/internet-shutdowns-deny-access-basic>.

²⁶ Bajoria, Jayshree. ““No Internet Means No Work, No Pay, No Food.”” *Human Rights Watch*, June 14, 2023.

<https://www.hrw.org/report/2023/06/14/no-internet-means-no-work-no-pay-no-food/internet-shutdowns-deny-access-basic>.

implementation of internet restrictions at a national level. It is the expression of a system which fuses socialist values with a capitalist system. The Chinese firewall serves the primary purpose of preventing information harmful the Chinese Communist Party (CCP) from entering the country²⁷ With almost 1 in 4 internet users residing in China, contrary to many models that predict enacting a kind of sovereign internet stifles economic growth, especially in the digital sphere, China sticks out like a sore thumb. China is number one in e-commerce, with digital retail sales in particular reaching volumes of over double that of the United States, which is often thought of as synonymous with freedom, and therefore an unrestricted internet. Names such as Alibaba or Tencent and Baidu are famous and even used outside of the country. Tencent is the developer of PUBG, one of the world's most played shooter video games, and Alibaba is known for its e-shopping services. It's only seeking to continue tightening its grasp on the internet now. Perhaps this is due to the reason that China's model isn't in fact that which would develop an intranet of sorts - similar to the above predicted RuNet. Rather, it is a firewall which is porous in the sense that users can access outside information within a certain extent. VPNs do function and permit users to bypass these limitations.²⁸ This way they strike the balance between exerting the control they seek and causing too much damage to the economy and upsetting their people.

The ways in which it might even have a positive impact on the Chinese economy are numerous. Importantly, the Chinese market is far too big for companies to ignore. The firewall isn't just a limitation, it is a declaration of the Chinese authorities that they are willing to control what goes in and out of the country. For foreign companies, that means that being denied access to the Chinese market would be detrimental to their profit. For example, six months after the launch of its iTunes and iBooks services, Apple was forced to discontinue them in China as a result of the authoritarian crackdown. Still, it announced a 1 billion dollar investment into a Chinese car company shortly after. Therefore these actions had two huge impacts. Firstly, they tackled a very important competitor to domestic services which would have otherwise been poised to dilute Chinese market share. This protected potentially smaller Chinese firms and made them more competitive compared to Apple which is very well established. Moreover, it is a strong move against the apps which are often shipped along with the devices from the factory which lead many to never even consider alternatives which would have certainly taken away from the Chinese market share. It also even had a measurable positive impact on the Chinese car firm - 1 billion dollars is no insignificant quantity. As a small case study, the evidence is overwhelmingly in favour of these measures.

²⁷ Denyer, Simon. "China's Scary Lesson to the World: Censoring the Internet Works." *The Washington Post*, May 23, 2016.

https://www.washingtonpost.com/world/asia_pacific/chinas-scary-lesson-to-the-world-censoring-the-internet-works/2016/05/23/413afe78-fff3-11e5-8bb1-f124a43f84dc_story.html.

²⁸ Denyer, Simon. "China's Scary Lesson to the World: Censoring the Internet Works." *The Washington Post*, May 23, 2016.

https://www.washingtonpost.com/world/asia_pacific/chinas-scary-lesson-to-the-world-censoring-the-internet-works/2016/05/23/413afe78-fff3-11e5-8bb1-f124a43f84dc_story.html.

However, there are undeniable drawbacks to such a method of control. The United States has formally designated the Great Chinese Firewall as a barrier to trade. Not everywhere in China are these controls so relaxed, impacting people far more significantly in the Western isolated regions where ethnic conflicts are most frequent. For example, in Chongqing, VPN users may be fined up to \$2178.²⁹ It's not just in the remote regions where pressure is felt however; Beijing employs 2 million people to police the people's thoughts where they don't accurately represent the regime's values. Such a massive army of people is bound to have a significant toll on the government's budget - there must be an immeasurable opportunity cost with the money and labour which is spent on censorship which could have otherwise been spent on funding startups, subsidizing companies or developing infrastructure which would have had an infinitely higher return on investment in terms of the economy. The sheer range of content censored by the Chinese government - not just political content but pornography, movie reviews, LGBTQ content, Western values or luxury, and general celebrity gossip.³⁰

This toll doesn't just affect the government's budget. Individual companies have to self-censor. This means that instead of focusing on creating content, more and more resources need to be diverted towards meeting ambiguous requirements.³¹ This ambiguity further reduces the incentive for creating start-ups particularly in the content-creation business because businesses take on risk with every material they produce. If they cannot be sure that the content won't be taken down, that risk can't accurately be judged and the investment may even be deemed to be not worthwhile. This is only one such example of the draining effect on the Chinese economy as innovation suffers and whole business models which otherwise thrive in most other places in the world, suffocate in China due to these restrictions. Due to the all-encompassing restrictions on communication, China also faces the risk of brain-drain, where brilliant people will seek to emigrate. Having services such as Google banned, scientists don't have access to the same tools as their western counterparts. They face the decision of staying at home, where they'll be far less competitive compared to the West, or choosing to emigrate to where they can carry their work unhampered by a lack of information.

Conclusion

²⁹ Globaltimes.cn. "Chongqing Regulations May Penalize VPN Users - Global Times," 2017. <https://www.globaltimes.cn/page/201703/1040014.shtml>.

³⁰ Balding, Christopher. "How Badly Is China's Great Firewall Hurting the Country's Economy?" Foreign Policy. Foreign Policy, July 18, 2017. <https://foreignpolicy.com/2017/07/18/how-badly-is-chinas-great-firewall-hurting-the-countrys-economy/>.

³¹ Balding, Christopher. "How Badly Is China's Great Firewall Hurting the Country's Economy?" Foreign Policy. Foreign Policy, July 18, 2017. <https://foreignpolicy.com/2017/07/18/how-badly-is-chinas-great-firewall-hurting-the-countrys-economy/>.

The economy is always most efficient when government intervention (or from other similar bodies) is minimal. For internet restrictions, this age old adage continues to seem to be correct. Internet restrictions are barriers to trade and therefore economic activity. Entire governments continue to believe that internet restrictions are an effective tool for their political agendas. This can only be justified by an uneducated understanding of the impact the internet really has. People continue to take the internet for granted, using it every day, across every task imaginable. From online banking, to filling out government forms, to collecting wages and everything in between. Internet plays a vital role in education - without which students would be unable to take exams or access online education³². It can affect people and their livelihoods as seen in India where people starve because they cannot receive their daily wages as a direct result of internet restrictions. The internet continues to be viewed as only an enabler of entertainment - a gimmick.

Even as economies develop towards offering more and more services and focusing less on other economic activities, the sense of urgency continues to go unnoticed. The scope of the internet continues to grow, and with it so do the consequences of switching it off. It acts as a great catalyst of exposure - particularly for tourism. Previously unexposed areas now thrive with their residents experiencing a dramatic rise in quality of life as a result of the hugely increased revenue within their community which can allow them to reinvest it in things such as infrastructure.³³ The internet does not only seek to fill already existing gaps - it has enabled entirely new business models which cannot hope to even exist without it. People such as streamers find themselves unemployed and unprepared to face life in the absence of the internet or even a suitable platform.

The only outlier continues to be China but that isn't unusual - China often seems to baffle westerners with their economic progress against its socialist background. However, just because it has done well for itself, does that mean that it is truly operating at full potential? Would it have experienced an even more radical rate of growth had it foregone restrictions altogether? Whilst on paper protectionist policies sound great, the reality is that the reduced competition will almost certainly have resulted in a less competitive China. The brain drain would also serve to only further reduce that competitiveness. The damages of this will be visible, however, only with time, as its effects cannot be immediate.

³² Bajoria, Jayshree. ““No Internet Means No Work, No Pay, No Food.”” *Human Rights Watch*, June 14, 2023.
<https://www.hrw.org/report/2023/06/14/no-internet-means-no-work-no-pay-no-food/internet-shutdowns-deny-access-basic>.

³³ Mitchell, Robert. “The Real Impact of Internet Shutdowns.” *Internet Society*, June 28, 2023.
<https://www.internetsociety.org/blog/2023/06/the-real-impact-of-internet-shutdowns/>.

These are only the tip of the iceberg of the economic impacts of internet restrictions. Even more hidden costs can include the uncompensated loss of the cost of labour.³⁴ Employees continue to have to be paid even during an internet outage as the terms of their wages are enforced by contract. This means that not only does a business lose on revenue, but it actively incurs losses. Similarly, besides the lost cost of labour, there are operating costs which may not be recovered. Things such as rent, subscriptions to various services, phone lines etc continue to have to be paid even when business is shallow. Then there is the reputation loss. A business is highly dependent on its reputation because it is one of the key factors that allow a business to attract clients. If it is unable to fulfill its obligations, even in light of a very solid excuse such as that of an internet blackout, its customers may continue to harbour resentment for any inconveniences. This may lead to customer satisfaction dropping and thereby alongside it so will its customer loyalty rate drop. This may be accompanied by a rise in sales of competitors in different countries or regions unaffected by the restrictions.

Therefore, the economic consequences of these restrictions are heaviest on those who impose and bear them. The government stands to forfeit revenue in the form of taxes in return for, supposedly, less civil unrest. People suffer the consequences of an absence of the internet in return for fewer job offerings and fewer opportunities to create their own enterprises. The domestic solutions which do survive are plagued by stagnation due to not being required to innovate nor having the same opportunities as foreign entities. Internet restrictions are a barrier to trade. By halting the free flow of information through the internet universally, the trend today of globalism would be reversed, with international trade and enterprise representing only a fraction of what it would have been previously.

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³⁴ www.pilotfiber.com. “The Hidden Costs of Internet Outages | Pilot Fiber.” Accessed November 4, 2024. <https://www.pilotfiber.com/blog/internet-outage-cost>.

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